

Bracket Sequences II

Time limit: 1.00 s

Memory limit: 512 MB

Your task is to calculate the number of valid bracket sequences of length n when a *prefix* of the sequence is given.

Input

The first input line has an integer n .

The second line has a string of k characters: the prefix of the sequence.

Output

Print the number of sequences modulo $10^9 + 7$.

Constraints

- $1 \leq k \leq n \leq 10^6$

Example

Input	Output
6 ((	2

Explanation: There are two possible sequences: $(()) ()$ and $((()))$.